

GIRIJANANDA CHOWDHURY UNIVERSITY
UT-1
Surveying & Geomatics (BCE402T)
Civil Engineering: Sem-4
2025
Full marks: 30

1*4=4

✓1. Fill in the blanks:

- i. The principle of chain surveying is _____.
- ii. Lines passing through points of equal declination are called _____.
- iii. The graphical method of adjustment of closing error is also known as _____.
- iv. The standard size of the plane table drawing board is _____.

2*5=10

✓2. Answer the following questions:

- i. Differentiate between Surveyor's compass and Prismatic compass.
- ii. Define Surveying, Whole Circle bearing, Agonic lines, Local attraction & Close traverse survey.

2*8=16

3. Solve the following:

- i. A base line was measured to be used 150m long with a tape at a field temperature of 27°C, the applied pull being 14kg. The tape was standardized at a temperature of 15°C with a pull of 8kg. If the designated length of the tape is 20m, weight of 1cm³ of tape is 7.86g, total weight of tape is 0.8kg. Take $E = 2.109 \times 10^6 \text{ kg/cm}^2$ and coefficient of thermal expansion of tape per degree Celsius = 11.2×10^{-6} , find the true length of the line.
- ii.
 - a) WCB of AB = 45°30'. Convert to QB.
 - b) QB of AB = S36°30'W. Convert to WCB.
 - c) If MB = S30°15'W. Find True bearing if Declination is 10°15' E.
 - d) If area on ground is 93,875 m² and area on map is 6.00cm X 6.25cm, find the scale of the map.
